

CHESAPEAKE MONEY POINT, VA, USA

WMO: 997305

Lat: 36.778N

Lon: 76.302W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.7	25.9	N/A	N/A	N/A	N/A	N/A	N/A	21.6	39.5	18.9	43.5	7.2	0	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	92.8	N/A	90.6	N/A	88.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.8	16.5	13.9	DB	17.8	96.6	4.6	2.7	14.5	98.6	11.7	100.2	9.1	101.7	5.8	103.7	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	42.1	44.0	50.0	61.0	68.8	76.9	80.4	79.3	73.8	64.2	53.2	47.9
	DBStd	15.40	10.06	9.46	9.33	7.94	6.81	5.07	3.55	3.38	4.89	7.36	7.74	9.27
	HDD50	836	284	215	121	8	0	0	0	0	3	53	152	
	HDD65	2979	709	590	472	168	41	1	0	0	2	103	359	534
	CDD50	5181	40	47	121	339	581	806	943	909	715	444	149	87
	CDD65	1851	0	2	7	49	158	357	478	444	267	79	5	5
	CDH74	15252	0	6	56	368	1078	3095	4703	3917	1642	367	9	11
	CDH80	5057	0	0	4	74	283	1123	1793	1308	409	63	0	0
Wind	WSAvg	6.2	6.1	6.8	7.6	7.3	6.8	6.1	5.8	5.1	5.2	5.8	6.1	5.9
Precipitation	PrecAvg	47.5	3.2	3.0	3.7	3.3	3.5	4.2	5.5	5.3	5.4	3.9	3.1	3.3
	PrecMax	60.4	5.9	8.6	9.9	6.0	5.3	10.4	9.4	13.6	17.5	8.1	11.0	6.8
	PrecMin	30.8	1.1	0.9	0.6	0.6	1.3	1.7	1.3	1.6	0.7	0.3	0.2	1.3
	PrecStd	8.1	1.3	1.6	1.7	1.3	1.0	2.0	2.0	2.9	3.9	2.1	2.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.9	75.2	79.7	86.2	90.5	95.0	96.3	94.2	90.5	85.5	75.5	75.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	67.6	69.8	74.5	82.0	86.3	91.8	93.1	91.2	87.3	81.4	72.2	71.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	63.3	66.1	70.0	78.5	83.5	89.1	90.8	89.0	84.8	78.1	68.9	67.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	59.0	59.9	65.9	74.6	80.2	86.4	88.4	86.7	82.1	75.2	65.9	63.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	14.6	15.4	15.7	16.8	15.3	14.3	13.9	12.6	12.1	13.9	14.1	14.3
		MCDBR	20.8	21.4	22.6	21.9	18.5	18.2	17.0	16.0	15.6	16.7	18.2	17.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.309	0.322	0.348	0.389	0.433	0.483	0.508	0.501	0.404	0.357	0.325	0.313	
	taud	2.535	2.494	2.449	2.347	2.255	2.169	2.113	2.118	2.381	2.505	2.543	2.553	
	Ebn at Noon	282	291	291	283	271	257	249	248	269	274	273	272	
	Edh at Noon	25	29	33	39	43	47	50	49	36	29	24	23	
All-Sky Solar Radiation	RadAvg	750	989	1318	1682	1889	2022	1925	1734	1415	1128	846	668	
	RadStd	61	106	103	123	159	127	94	101	112	129	68	53	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+299	+189

Nomenclature: See separate page